

HRITHIK NANDI

PERSONAL DATA

DATE OF BIRTH: 18 March 2000
ADDRESS (WORK): Institute for Advancing Intelligence TCG CREST, Kolkata, India
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RESEARCH POSITIONS HELD

Aug. 2022 – Present | **Cryptology and Security Group**, IAI TCG CREST, Kolkata, India
Position: Senior Research Fellow

EDUCATION

Aug. 2022 – Present | **Ph.D. in Computer Science**
Institute for Advancing Intelligence TCG CREST
Ramakrishna Mission Vivekananda Educational and Research Institute
Advisor(s): Dr. Avijit Dutta & Dr. Soumitra Samanta

Aug. 2020 – Jul. 2022 | **M.Sc. in Mathematics**
Ramakrishna Mission Vivekananda Educational and Research Institute

Jul. 2017 – Jun. 2020 | **B.Sc. in Mathematics**
Ramakrishna Mission Vivekananda Centenary College

RESEARCH INTERESTS

- Provable Security
- Design and Analysis of Symmetric Key Cryptographic Schemes
- Post-quantum Security of Symmetric Key Cryptographic Schemes

PUBLICATIONS

1. Nilanjan Datta, Avijit Dutta, Sougata Mandal, Hrithik Nandi, Amlan Sinha. **Post-Quantum Security of Keyed Sum of Permutations and Its Siblings**. *IACR ASIACRYPT 2026*. [DOI, [ePrint](#)]
2. Nilanjan Datta, Hrithik Nandi, Soumit Pal, Yu Sasaki, Patrick Struck, Maximiliane Weishäupl. **Generic Committing Attacks: Zero-Padded Ascon is Less Secure than Expected**. *IACR CRYPTO 2026*. [DOI, [ePrint](#)]
3. Nilanjan Datta, Jean Paul Degabriele, Avijit Dutta, Vukašin Karadžić, Hrithik Nandi. **Rugged Pseudorandom Permutations with Beyond-Birthday-Bound Security**. *ACM AsiaCCS 2026*. [DOI, [ePrint](#)]
4. Ritam Bhaumik, Nilanjan Datta, Avijit Dutta, Ashwin Jha, Sougata Mandal, Bart Mennink, Hrithik Nandi, Yaobin Shen. **How to Build a Short-Input Random Oracle from Public Random Permutations**. *IACR EUROCRYPT 2026*. [DOI, [ePrint](#)]
5. Nilanjan Datta, Avijit Dutta, Shibam Ghosh, Eik List, Hrithik Nandi. **HCTR+: An Optimally Secure TBC-based Accordion Mode**. *IACR ToSC 2025(3)*. [DOI, [ePrint](#)]

6. Nilanjan Datta, Avijit Dutta, Sougata Mandal, Hrithik Nandi. **Sequential Indifferentiability of STH and EDM**. *IACR CiC 2025(2)*. [[DOI](#), [ePrint](#)]
7. Ritam Bhaumik, Wonseok Choi, Avijit Dutta, Cuauhtemoc Mancillas López, Hrithik Nandi, Yaobin Shen. **Efficient Variants of TNT with BBB Security**. *ProvSec 2024*. [[DOI](#), [ePrint](#)]

PREPRINTS

1. Sougata Mandal, Hrithik Nandi, Amlan Sinha. **Committing Security of BBB Secure MACs**. [[ePrint](#)]
2. Avik Chakraborti, Bishwajit Chakraborty, Nilanjan Datta, Avijit Dutta, Ashwin Jha, Sougata Mandal, Hrithik Nandi, Mridul Nandi, Abishanka Saha. **Naor-Reingold goes Beyond-the-Birthday-Bound**. [[ePrint](#)]

INVITED AND CONTRIBUTED TALKS

- | | |
|-----------|--|
| Mar. 2026 | HCTR+: An Optimally Secure TBC-based Accordion Mode
FSE 2026, Singapore |
| Mar. 2026 | Short-Input Random Oracle from Public Random Permutations
ASK 2026, NTU, Singapore |
| Dec. 2025 | Sequential Indifferentiability of STH and EDM
Crypto Winter School 2025, IIT Bhilai, India |
| Dec. 2024 | Optimally Secure TBC Based Accordion Mode
ASK 2024, TCG CREST, Kolkata, India |
| Sep. 2024 | Efficient Variants of TNT with BBB Security
ProvSec 2024, Griffith University, Gold Coast, Australia |

WORKSHOPS AND CONFERENCES ATTENDED

- | | |
|--------------------|--|
| Mar. 23 – 27, 2026 | Fast Software Encryption (FSE 2026)
Singapore, Singapore |
| Mar. 19 – 22, 2026 | Asian-workshop on Symmetric Key Cryptography (ASK 2026)
NTU, Singapore |
| Sep. 01 – 05, 2025 | Workshop on Generic Attacks and Proofs in Symmetric Cryptography
(GAPS 2025), NTU, Singapore |
| Dec. 14 – 17, 2024 | Asian-workshop on Symmetric Key Cryptography (ASK 2024)
Kolkata, India |
| Dec. 09 – 13, 2024 | Asiacrypt 2024
Kolkata, India |
| Sep. 25 – 27, 2024 | International Conference on Provable and Practical Security
(ProvSec 2024), Gold Coast, Australia |
| Dec. 10 – 13, 2023 | Indocrypt 2023
Goa, India |
| Dec. 11 – 14, 2022 | Indocrypt 2022
Kolkata, India |

Sub-reviewer | CRYPTO 2026

Teaching Assistant	Discrete Mathematics, IAI TCG CREST, Aug. 2024 – Dec. 2024
	Cryptology, IAI TCG CREST, Aug. 2023 – Dec. 2023

REFERENCES

Dr. Avijit Dutta
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Institute for Advancing Intelligence, TCG CREST, Kolkata, India
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Dr. Nilanjan Datta
Associate Professor
Institute for Advancing Intelligence, TCG CREST, Kolkata, India
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DECLARATION

I hereby declare that the information stated above is true to the best of my knowledge and belief.

Place: Kolkata, India

Hrithik Nandi